

## GRIMSEL-HOSPIZ, Switzerland

WMO: 067440

Lat: 46.5717N

Lon: 8.3333E

Elev: 6447

StdP: 11.58

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 3.3        | 7.1 | -15.0                         | 3.1 | 15.6 | -10.4 | 4.0 | 16.5 | 42.1                  | 25.8 | 37.5 | 28.7 | 8.6                   | 350  | 0.847 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.8                   | 66.4            | 53.5 | 63.9 | 52.3 | 61.4 | 51.3 | 55.2                | 62.8 | 53.8 | 61.4 | 52.5 | 59.6 | 10.7                 | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 52.4                            | 74.6 | 57.2 | 50.7 | 69.9 | 56.0 | 49.2 | 65.9 | 55.0 | 26.1          | 62.9 | 25.1 | 61.6 | 24.2 | 59.9 | 61.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 29.9          | 25.7 | 22.9 | DB                         | -1.2 | 71.2               | 5.9 | 2.2                                                | -5.4 | 72.7       | -8.9 | 74.0       | -12.2 | 75.2       | -16.5 | 76.8 | (4) |
| (5)               |      |      | WB                         | -2.5 | 57.8               | 5.6 | 1.8                                                | -6.5 | 59.1       | -9.7 | 60.1       | -12.9 | 61.1       | -16.9 | 62.4 | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 36.7   | 23.3  | 22.5  | 27.3  | 33.6  | 39.8  | 48.4  | 51.2  | 50.9  | 45.0  | 40.7  | 31.5  | 25.7 |
|                                                                               | DBStd       | 12.66  | 7.35  | 8.71  | 7.61  | 6.91  | 7.10  | 7.41  | 6.44  | 6.69  | 6.86  | 7.72  | 8.01  | 7.73 |
|                                                                               | HDD50       | 5150   | 829   | 771   | 702   | 492   | 324   | 117   | 66    | 72    | 173   | 295   | 557   | 752  |
|                                                                               | HDD65       | 10316  | 1294  | 1191  | 1167  | 942   | 782   | 500   | 428   | 436   | 599   | 754   | 1006  | 1217 |
|                                                                               | CDD50       | 309    | 0     | 0     | 0     | 0     | 7     | 68    | 102   | 101   | 24    | 7     | 0     | 0    |
|                                                                               | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDH74       | 1      | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 11.8   | 13.3  | 13.1  | 12.4  | 11.9  | 11.1  | 10.3  | 10.2  | 10.7  | 10.3  | 12.2  | 12.9  | 13.1 |
| Precipitation                                                                 | PrecAvg     | 62.2   | 3.9   | 4.1   | 4.1   | 4.9   | 6.3   | 6.1   | 5.5   | 6.1   | 5.4   | 5.0   | 6.3   | 4.5  |
|                                                                               | PrecMax     | 80.4   | 9.5   | 11.6  | 8.0   | 9.1   | 14.1  | 10.2  | 10.0  | 10.8  | 13.3  | 11.9  | 19.4  | 11.5 |
|                                                                               | PrecMin     | 43.5   | 0.9   | 0.8   | 0.7   | 0.8   | 1.8   | 2.4   | 3.0   | 1.9   | 1.6   | 0.6   | 1.2   | 0.5  |
|                                                                               | PrecStd     | 8.2    | 2.3   | 2.5   | 2.1   | 2.2   | 3.1   | 1.9   | 1.8   | 1.9   | 2.8   | 2.7   | 4.0   | 2.4  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 41.3  | 43.8  | 45.5  | 52.6  | 61.5  | 70.3  | 69.2  | 69.9  | 62.8  | 59.1  | 49.8  | 42.5 |
|                                                                               |             | MCWB   | 32.1  | 32.4  | 34.4  | 41.2  | 48.6  | 53.4  | 55.1  | 53.5  | 50.6  | 43.8  | 38.1  | 32.4 |
|                                                                               | 2%          | DB     | 37.8  | 39.7  | 42.1  | 48.4  | 55.8  | 65.8  | 65.6  | 66.1  | 59.3  | 55.0  | 46.6  | 39.6 |
|                                                                               |             | MCWB   | 30.4  | 29.7  | 32.5  | 39.0  | 45.4  | 52.7  | 53.8  | 53.4  | 48.9  | 43.0  | 36.7  | 30.2 |
|                                                                               | 5%          | DB     | 35.4  | 36.0  | 39.6  | 45.5  | 52.5  | 62.5  | 63.2  | 63.0  | 56.4  | 52.0  | 43.7  | 37.4 |
|                                                                               |             | MCWB   | 28.8  | 28.4  | 31.6  | 37.3  | 43.8  | 51.7  | 52.6  | 51.9  | 47.4  | 41.9  | 34.7  | 29.6 |
|                                                                               | 10%         | DB     | 33.0  | 32.9  | 37.2  | 42.8  | 49.5  | 59.1  | 60.8  | 60.4  | 54.0  | 49.6  | 41.2  | 35.5 |
|                                                                               |             | MCWB   | 28.1  | 27.3  | 30.6  | 35.9  | 42.1  | 50.2  | 51.7  | 51.0  | 46.3  | 40.9  | 34.1  | 29.1 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 35.6  | 34.5  | 36.7  | 42.8  | 50.1  | 56.6  | 57.0  | 57.1  | 52.8  | 47.0  | 41.2  | 34.9 |
|                                                                               |             | MCDB   | 38.1  | 39.6  | 41.2  | 49.5  | 58.1  | 62.9  | 64.8  | 63.9  | 58.2  | 54.2  | 47.3  | 38.4 |
|                                                                               | 2%          | WB     | 32.7  | 31.9  | 34.7  | 40.3  | 46.9  | 54.5  | 55.2  | 54.8  | 50.8  | 45.3  | 38.4  | 33.4 |
|                                                                               |             | MCDB   | 35.2  | 35.6  | 39.3  | 46.7  | 54.0  | 62.4  | 62.8  | 62.4  | 56.8  | 51.1  | 43.2  | 37.2 |
|                                                                               | 5%          | WB     | 30.6  | 30.0  | 33.1  | 38.4  | 44.7  | 52.8  | 53.9  | 53.3  | 49.0  | 44.1  | 36.7  | 31.8 |
|                                                                               |             | MCDB   | 33.8  | 34.7  | 37.5  | 44.1  | 51.1  | 61.0  | 61.2  | 60.7  | 54.5  | 49.4  | 41.5  | 35.3 |
|                                                                               | 10%         | WB     | 28.8  | 28.0  | 31.6  | 36.6  | 42.9  | 50.8  | 52.4  | 52.0  | 47.4  | 42.7  | 35.2  | 30.1 |
|                                                                               |             | MCDB   | 32.2  | 32.7  | 35.6  | 41.7  | 48.5  | 58.2  | 59.3  | 58.7  | 52.3  | 48.1  | 40.1  | 33.9 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 8.6   | 9.2   | 9.7   | 9.5   | 10.1  | 11.3  | 10.8  | 10.4  | 9.5   | 9.3   | 7.9   | 8.4  |
|                                                                               |             | MCDBR  | 9.9   | 11.1  | 11.5  | 11.9  | 13.6  | 15.6  | 13.8  | 13.2  | 12.3  | 12.5  | 10.2  | 9.3  |
|                                                                               |             | MCWBR  | 6.7   | 6.8   | 7.6   | 8.1   | 8.8   | 10.3  | 9.3   | 8.9   | 8.1   | 7.0   | 5.7   | 6.0  |
|                                                                               | 5% WB       | MCDBR  | 9.1   | 10.5  | 10.0  | 11.1  | 12.8  | 14.4  | 13.0  | 12.3  | 11.1  | 10.5  | 8.7   | 8.8  |
|                                                                               |             | MCWBR  | 7.5   | 8.3   | 7.2   | 7.8   | 8.6   | 10.1  | 9.5   | 9.0   | 8.0   | 6.7   | 5.7   | 6.8  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.229  | 0.253 | 0.294 | 0.337 | 0.354 | 0.361 | 0.359 | 0.349 | 0.325 | 0.299 | 0.260 | 0.227 |      |
|                                                                               | taud        | 2.339  | 2.283 | 2.233 | 2.227 | 2.286 | 2.313 | 2.311 | 2.356 | 2.417 | 2.470 | 2.454 | 2.370 |      |
|                                                                               | Ebn at Noon | 284    | 295   | 295   | 291   | 289   | 286   | 285   | 284   | 282   | 275   | 269   | 272   |      |
|                                                                               | Edh at Noon | 23     | 30    | 37    | 41    | 41    | 40    | 39    | 36    | 31    | 25    | 20    | 19    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 490    | 791   | 1171  | 1409  | 1528  | 1799  | 1789  | 1554  | 1210  | 799   | 485   | 407   |      |
|                                                                               | RadStd      | 60     | 96    | 169   | 223   | 150   | 162   | 156   | 140   | 117   | 92    | 71    | 55    |      |

## Historical Trends

| DBAvg                    | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|--------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                          | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)                      | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only             | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (118 neighbors) | N/A     | N/A    | +1.31   | +0.76 | +0.58 | N/A         | N/A   | +104  | +29   |

Nomenclature: See separate page